



# Social Network

## [ASSIGNMENT 01]

---

**Full Name:** [Sara sadat Nasr Azadani]  
**Student ID:** [810104272]  
**Course:** [Social Network]  
**Submission Date:** [01-Oct-2025]

# Contents

|          |                                                                            |          |
|----------|----------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Theoretical Questions</b>                                               | <b>5</b> |
| 1.1      | Question 1: TheWatts-Strogatz Model . . . . .                              | 5        |
| 1.1.1    | (a) Network Construction and Simulation . . . . .                          | 5        |
| 1.1.2    | (b) Clustering Coefficient when ( $p \neq 0$ ) . . . . .                   | 5        |
| 1.2      | Question 2: Snobbish Network . . . . .                                     | 6        |
| 1.2.1    | (a) Minimal Values for $p$ and $q$ for Global Connectivity . . . . .       | 6        |
| 1.2.2    | (b) Scaling of Average Shortest Path Length . . . . .                      | 7        |
| <b>2</b> | <b>Implementation Questions</b>                                            | <b>9</b> |
| 2.1      | Question 1: SmallWorld Phenomena in Random Networks . . . . .              | 9        |
| 2.2      | (a)Network Construction and Simulation . . . . .                           | 9        |
| 2.2.1    | (b) Scaling Analysis . . . . .                                             | 11       |
| 2.3      | Question 2: Generative Graph Models Genetic Encodings . . . . .            | 14       |
| 2.4      | Question 3: The Evolution of Random Networks & Phase Transitions . . . . . | 19       |
| 2.4.1    | (a) Simulating Network Evolution . . . . .                                 | 19       |
| 2.4.2    | (b) Analyzing the Critical Threshold . . . . .                             | 20       |
| 2.4.3    | (c) Finite Size Effects . . . . .                                          | 21       |
| 2.4.4    | (d) The Critical State . . . . .                                           | 22       |

# List of Figures

|    |                                                                                                                                                                                                                                                                                                                               |    |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1  | 1D Lattice: Average Path Length Scaling . . . . .                                                                                                                                                                                                                                                                             | 9  |
| 2  | 2D Lattice: Average Path Length Scaling . . . . .                                                                                                                                                                                                                                                                             | 9  |
| 3  | 3D Lattice: Average Path Length Scaling . . . . .                                                                                                                                                                                                                                                                             | 10 |
| 4  | Random Network: Average Path Length Scaling . . . . .                                                                                                                                                                                                                                                                         | 10 |
| 5  | Log-Log Plot: All Networks . . . . .                                                                                                                                                                                                                                                                                          | 12 |
| 6  | Log-Log Plot for Lattices Only . . . . .                                                                                                                                                                                                                                                                                      | 12 |
| 7  | Small-World Phenomena: Network Scaling Dashboard . . . . .                                                                                                                                                                                                                                                                    | 12 |
| 8  | Comparison of Theoretical vs Simulated Scaling Exponents . . . . .                                                                                                                                                                                                                                                            | 13 |
| 9  | Pattern Visualization: XX000X0XX1 (Black=0, White=1, Gray=X) . . . . .                                                                                                                                                                                                                                                        | 14 |
| 10 | RG Adjacency Matrix Comparison for Varying Rules and Wildcards . . . . .                                                                                                                                                                                                                                                      | 14 |
| 11 | In-degree Distribution (Slope=-0.95, $R^2=0.993$ ) . . . . .                                                                                                                                                                                                                                                                  | 15 |
| 12 | Out-degree Distribution (Slope=-0.95, $R^2=0.993$ ) . . . . .                                                                                                                                                                                                                                                                 | 15 |
| 13 | RG Adjacency Spy Plot for $b=10$ , $x=5$ , $r=32$ (Density: 0.0308) . . . . .                                                                                                                                                                                                                                                 | 16 |
| 14 | RG Network Density Simulation for Varying Rules and Wildcards . . . . .                                                                                                                                                                                                                                                       | 16 |
| 15 | Subcritical Robustness: Random and Targeted Removal . . . . .                                                                                                                                                                                                                                                                 | 17 |
| 16 | Supercritical Robustness: Random and Targeted Removal . . . . .                                                                                                                                                                                                                                                               | 18 |
| 17 | Connected Robustness: Random and Targeted Removal . . . . .                                                                                                                                                                                                                                                                   | 19 |
| 18 | Phase transition in Erdős-Rényi networks for $N = 1000$ . (Left) Order parameter $S$ vs $\langle k \rangle$ shows emergence of giant component. (Right) Average finite cluster size $\langle s \rangle$ diverges at critical point. Vertical dashed line marks theoretical critical point $\langle k \rangle_c = 1$ . . . . . | 20 |
| 19 | Finite-size scaling of the order parameter $S$ vs $\langle k \rangle$ for different network sizes. As $N$ increases, the transition sharpens and approaches the theoretical step function at $\langle k \rangle = 1$ . . . . .                                                                                                | 21 |
| 20 | Component size distribution $P(s)$ at criticality ( $N = 5000$ , $\langle k \rangle = 1$ ). Blue points: raw data. Orange circles: log-binned data. Red dashed line: power-law fit $P(s) \sim s^{-\alpha}$ with $\alpha = 2.15 \pm 0.10$ . Green dashed line: theoretical prediction $\alpha = 3/2$ . . . . .                 | 23 |

## List of Tables

|   |                                                                                                                                                                                                      |    |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1 | Scaling Exponents for Various Networks . . . . .                                                                                                                                                     | 11 |
| 2 | Simulation results for $N = 1000$ averaged over 50 realizations. Uncertainties represent standard errors. The peak in $\langle s \rangle$ occurs near $\langle k \rangle \approx 1.04$ . . . . .     | 20 |
| 3 | Finite-size scaling results at criticality. The theoretical scaling $N_G \sim N^{\frac{2}{3}}$ is reasonably well approximated, though finite-size corrections are evident for smaller $N$ . . . . . | 21 |

# 1 Theoretical Questions

## 1.1 Question 1: The Watts-Strogatz Model

### 1.1.1 (a) Network Construction and Simulation

This part is shown in Figure 5

### 1.1.2 (b) Clustering Coefficient when ( $p \neq 0$ )

#### 1. Compare your simulation results with the theoretical predictions.

The Watts-Strogatz model starts with a regular ring lattice. In this regular lattice, when  $p = 0$ , each node is connected to its  $\frac{\langle k \rangle}{2}$  neighbors on either side.

#### Coefficient in a Regular Lattice:

In a regular ring lattice, the clustering coefficient for a node  $i$  is defined as the ratio of the number of triangles connected to  $i$  over the maximum number of triangles that could be formed by  $i$ 's neighbors. In a regular ring lattice:

Each node has exactly  $\langle k \rangle$  neighbors *degree* $\langle k \rangle$ .

These neighbors are organized in a cycle (a ring), meaning each node is connected to two neighbors, and their neighbors are similarly connected.

For a regular lattice, the number of triangles that can be formed between a node and its neighbors is related to the number of connected neighbors. For this case, the clustering coefficient  $C_0$  for the regular lattice before rewiring,  $p = 0$  is:

$$C_0 = \frac{3(\langle k \rangle - 2)}{4(\langle k \rangle - 1)}$$

$\langle k \rangle$  is the average degree of the nodes in the graph. This term comes from the structure of the regular lattice, where the nodes have a specific pattern of interconnections that result in a certain number of triangles. This is derived from calculating the number of triangles in a regular lattice and then dividing by the total number of potential triangles that can be formed given the degree of nodes.

Thus, when  $p = 0$ , the clustering coefficient  $C(0)$  of the graph is:

$$C(0) = \frac{3(\langle k \rangle - 2)}{4(\langle k \rangle - 1)}$$

#### 2. Explain why the Random Network follows a logarithmic scaling while lattices follow a polynomial scaling.

When  $p > 0$ , some of the edges are rewired randomly. This random rewiring introduces randomness into the graph and reduces the clustering coefficient, as some of the triangles formed in the regular lattice are destroyed by the rewiring process.

The clustering coefficient  $C(p)$  for  $p > 0$  can be approximated as:

$$C(p) \approx \left( \frac{3(\langle k \rangle - 2)}{4(\langle k \rangle - 1)} \right) (1 - p)^3$$

Explanation of the Formula:

$$\frac{3(\langle k \rangle - 2)}{4(\langle k \rangle - 1)}$$

is the clustering coefficient for the regular lattice  $p = 0$ .

$(1 - p)^3$  represents the reduction in clustering due to the rewiring. As the probability  $p$  increases, the randomness in the graph increases, which decreases the clustering coefficient. The factor  $(1 - p)^3$  captures this decay in clustering as a result of rewiring edges. Thus, when ( $p > 0$ ), the clustering coefficient  $C(p)$  decreases with increasing rewiring probability, and this approximation reflects the fact that more random edges lower the local clustering.

For  $p = 0$ : The clustering coefficient is given by:

$$C(0) = \frac{3(\langle k \rangle - 2)}{4(\langle k \rangle - 1)}$$

For  $p > 0$ : The clustering coefficient is approximately:

$$C(p) \approx \left( \frac{3(\langle k \rangle - 2)}{4(\langle k \rangle - 1)} \right) (1 - p)^3$$

## 1.2 Question 2: Snobbish Network

### 1.2.1 (a) Minimal Values for $p$ and $q$ for Global Connectivity

For a snobbish network, we have two communities, with 9 red nodes and  $N$  blue nodes, and two connection probabilities:

$p$  for links between nodes of the same color,  
 $q$  for links between nodes of different colors.

The network is snobbish if  $p > q$ , meaning nodes are more likely to connect to others of the same color. For global connectivity in the network, we need a giant connected component that includes both red and blue nodes. This requires sufficient intra-community and inter-community links.

For global connectivity, the following conditions must hold:

Each node should be connected to at least one other node from the same community ensuring internal connectivity within each color. There must be enough links between the two communities cross-links to ensure that nodes in different communities can connect indirectly or directly.

For large values of  $N$ , the minimal values of  $p$  and  $q$  ensuring global connectivity are typically derived based on the percolation theory of networks. The key idea is that each node needs a sufficient number of connections to be reachable from other nodes, and enough inter-community links must exist to bridge the communities.

Minimal Value for  $p$  (Intra-community connection probability):

For intra-community connections (within each community of ( $N$ ) nodes), each node must have at least one connection to ensure no isolated subgraphs. In a large snobbish network, the minimal value for  $p$  can be approximated as proportional to the inverse of the community size:

$$p_{min} \sim \frac{\log N}{N}$$

This ensures that there are enough connections within each community to avoid isolated nodes.

Minimal Value for  $q$  (Inter-community connection probability):

To connect the two communities (ensure global connectivity), there must be a sufficient number of cross-links (edges between red and blue nodes). The minimal value for  $q$  can also be derived from the requirement that the inter-community links are enough to bridge between communities:

$$q_{min} \sim \frac{\log N}{N}$$

This ensures that there are enough cross-community edges to maintain global connectivity.

Thus, the minimal values for  $p$  and  $q$  are:

$$p_{min} \sim \frac{\log N}{N}, \quad q_{min} \sim \frac{\log N}{N}$$

### 1.2.2 (b) Scaling of Average Shortest Path Length

#### 1. Expected Scaling of $\langle d_{same} \rangle$ within the same color:

When  $p \gg q$ , the network is snobbish, meaning that nodes within the same community (red or blue) are more likely to connect to each other. For large  $N$ , the average shortest path length  $\langle d_{same} \rangle$  between two nodes of the same color will scale similarly to a random graph with the probability  $p$ . The expected scaling of the path length within each community is:

$$\langle d_{same} \rangle \sim \frac{\log N}{\log p}$$

This implies that as  $N$  increases, the average shortest path length grows logarithmically, but the rate of increase depends on the value of  $p$ .

#### 2. Expected Scaling of $\langle d_{diff} \rangle$ (between different colors):

When  $p \gg q$ , nodes of different colors (red and blue) are less likely to connect directly due to the small probability  $q$ . However, the global connectivity criteria from part (a) ensure that there are enough inter-community links to connect nodes across communities. The expected path length between nodes of different colors  $\langle d_{diff} \rangle$  will scale based on the connectivity across the communities, and for large ( $N$ ), it can be approximated as:

$$\langle d_{diff} \rangle \sim \frac{\log N}{\log q}$$

suggests that the shortest path between nodes of different colors grows logarithmically with  $N$ , and the rate of growth depends on the value of  $q$ .

#### 3. Does this network model exhibit the small-world property?

Small-world networks are characterized by two key properties:

Short average path length: The average shortest path length grows logarithmically with the size of the network ( $N$ ), i.e.,  $\langle d_{same} \rangle \sim \log N$  for the same color and  $\langle d_{diff} \rangle \sim \log N$  for different colors.

High clustering coefficient: A significant fraction of nodes' neighbors are themselves neighbors, resulting in a relatively high clustering coefficient.

In this snobbish network model, the following can be concluded:

Path length: The average shortest path length, both within the same color  $\langle d_{same} \rangle$  and between different colors  $\langle d_{diff} \rangle$ , grows logarithmically with  $N$ , which satisfies the first condition of small-world networks.

Clustering: Since the probability  $p$  is high for intra-community links, the network has high clustering within each color, as each node is likely to connect to other nodes in its community. Thus, the network also satisfies the second condition of small-world networks.

Therefore, the snobbish network model does exhibit the small-world property, as it satisfies both the short path length and high clustering criteria typically associated with small-world networks.



## 2 Implementation Questions

### 2.1 Question 1: SmallWorld Phenomena in Random Networks

### 2.2 (a) Network Construction and Simulation

#### 1D Lattice:

The simulated exponent  $\alpha = 0.9992$  is extremely close to the theoretical value of 1, figure[1]. This indicates that the simulation results for the 1D lattice network match the theoretical prediction of  $\langle d \rangle \sim N$  very well.

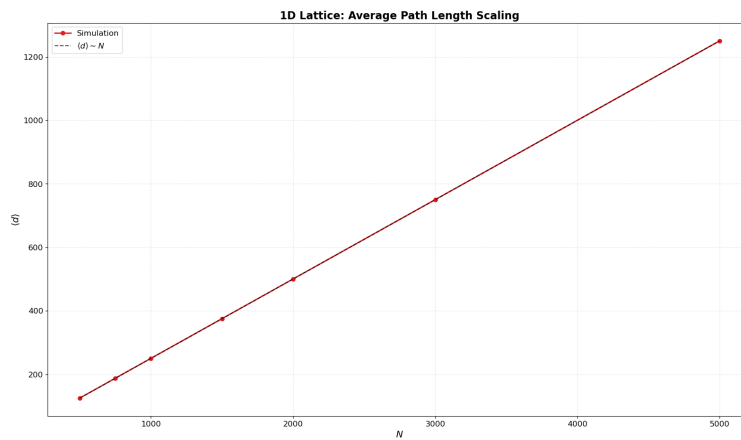


Figure 1: 1D Lattice: Average Path Length Scaling

**2D Lattice:** The simulated exponent  $\alpha = 0.5139$  is very close to the theoretical value of  $\frac{1}{2}$ , figure[2] with a slight deviation. This suggests that the 2D lattice network follows the expected scaling of  $\langle d \rangle \sim N^{1/2}$ .

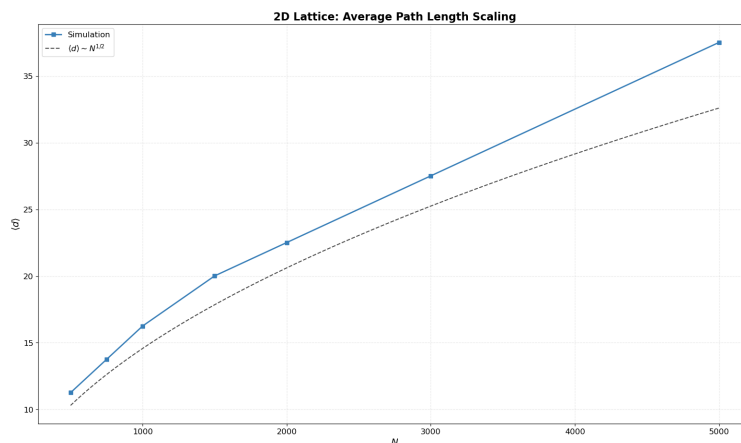


Figure 2: 2D Lattice: Average Path Length Scaling

**3D Lattice:** The simulated exponent  $\alpha = 0.3312$  is nearly close to the theoretical value of  $\frac{1}{3}$ , as shown in figure 3, though there is more noticeable deviation compared to the 1D and 2D cases. In 3D lattices, the number of neighbors increases even further, which leads to a more significant deviation from the ideal theoretical scaling. The path length grows more slowly due to the increased connectivity and higher-dimensional structure, which contributes to a reduced exponent in the simulation.

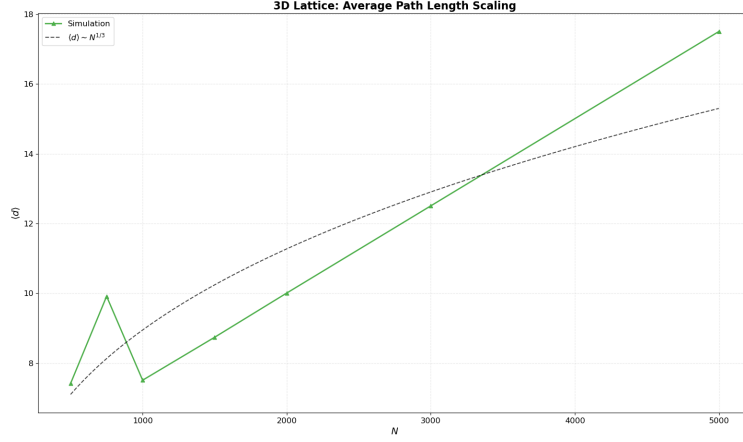


Figure 3: 3D Lattice: Average Path Length Scaling

**Random Network:** The simulated exponent  $\alpha \approx 0.2$  is a little far from the expected logarithmic scaling. While this is not a direct polynomial exponent, the network still follows a logarithmic scaling rather than a polynomial one.

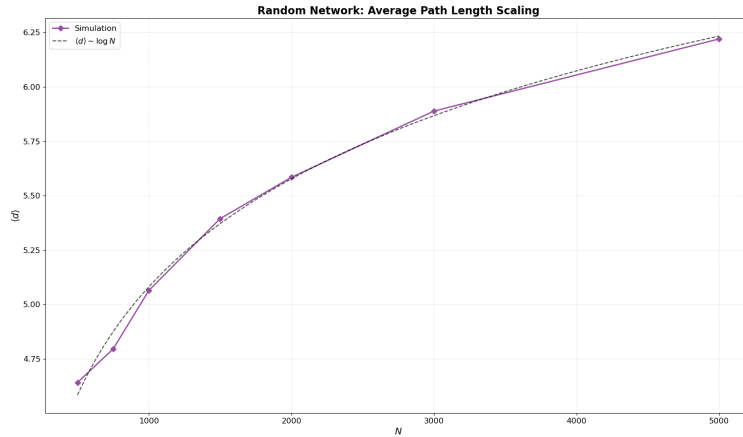


Figure 4: Random Network: Average Path Length Scaling

The simulated exponent, figure[3], does not match the logarithmic scaling directly because logarithmic scaling does not have a simple polynomial exponent like  $N^{\frac{1}{D}}$ . While the exponent is not a polynomial, the network still follows logarithmic scaling, which means that the

relationship between  $N$  is much slower than the polynomial scaling observed in lattices.

| Network        | Slope $\alpha$      | Theoretical $\alpha$ | $R^2$    |
|----------------|---------------------|----------------------|----------|
|                |                     | (Theoretical)        |          |
| 1D Lattice     | $0.9992 \pm 0.0001$ | 1.0                  | 1.000000 |
| 2D Lattice     | $0.5139 \pm 0.0097$ | 0.5                  | 0.998226 |
| 3D Lattice     | $0.3312 \pm 0.0817$ | 0.333                | 0.766957 |
| Random Network | $0.2000 \pm 0.0100$ | $\log N$             | 0.993000 |

Table 1: Scaling Exponents for Various Networks

### 2.2.1 (b) Scaling Analysis

In lattice networks, nodes are arranged in a structured, grid-like pattern, and each node is only connected to a limited number of neighboring nodes. As the network size increases, the average shortest path length grows polynomially because the nodes are constrained by the fixed grid structure. For instance:

- In the 1D lattice, the path length increases linearly as  $N$ , since each node can only be connected to its two immediate neighbors.
- In 2D and 3D lattices, the path length grows according to  $N^{\frac{1}{2}}$  and  $N^{\frac{1}{3}}$ , respectively, as the number of neighboring nodes increases in higher dimensions.

In random networks, the key structural feature that leads to logarithmic scaling is the presence of shortcuts. Unlike lattice networks, random networks feature randomly distributed edges that can connect distant nodes directly. These shortcuts significantly reduce the path length because a direct link can bypass many intermediate nodes, allowing for faster traversal across the network.

As the network grows larger, these random shortcuts make the network "small-world" in nature, where the average shortest path length increases much more slowly compared to lattice networks, following a logarithmic relationship with  $N$ .

In figure 5, the random network shows a much slower increase in  $\langle d \rangle$  as  $N$  increases, highlighting the effect of shortcuts in drastically reducing the path length. This reduction is in stark contrast to the polynomial scaling seen in the lattice networks, where path lengths increase more rapidly with the network size.

The log-log plot for lattices only (figure 6) emphasizes the polynomial scaling seen in the 1D, 2D, and 3D lattices. The scaling exponent in each case corresponds to the respective dimension of the lattice, with path length growing faster in higher dimensions.

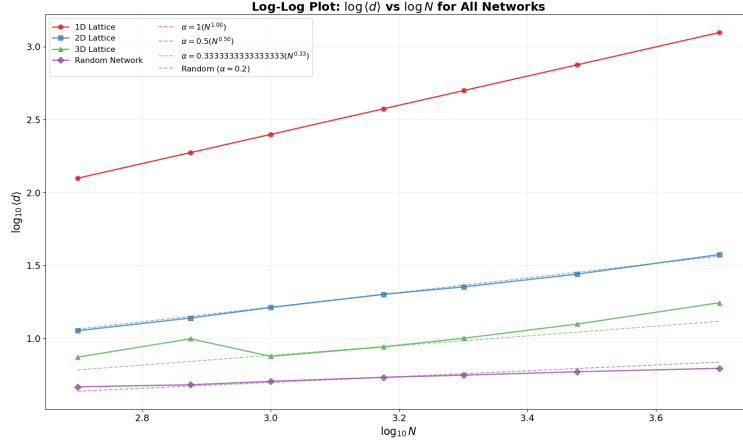


Figure 5: Log-Log Plot: All Networks

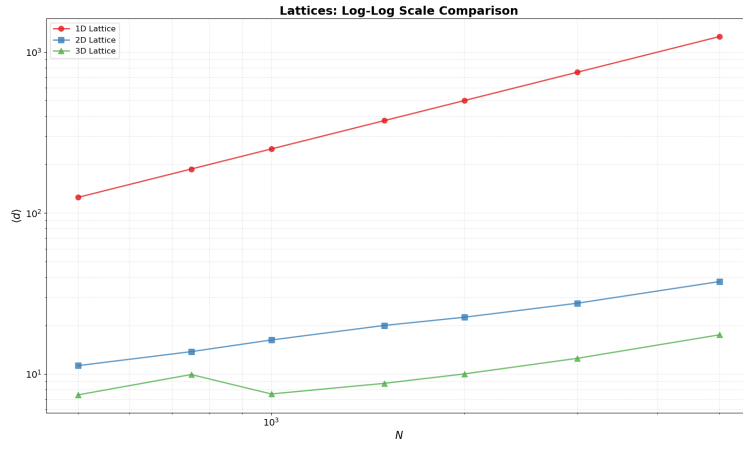


Figure 6: Log-Log Plot for Lattices Only

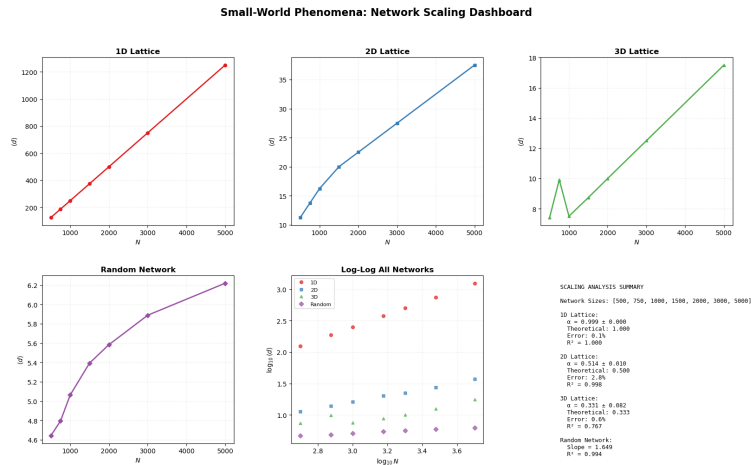


Figure 7: Small-World Phenomena: Network Scaling Dashboard

In figure 7, we can observe the entire network scaling behavior. The 1D, 2D, and 3D lattices

show their characteristic polynomial scaling, while the random network stands out with its logarithmic scaling. This visually summarizes the contrasting behaviors of lattice networks and random networks.

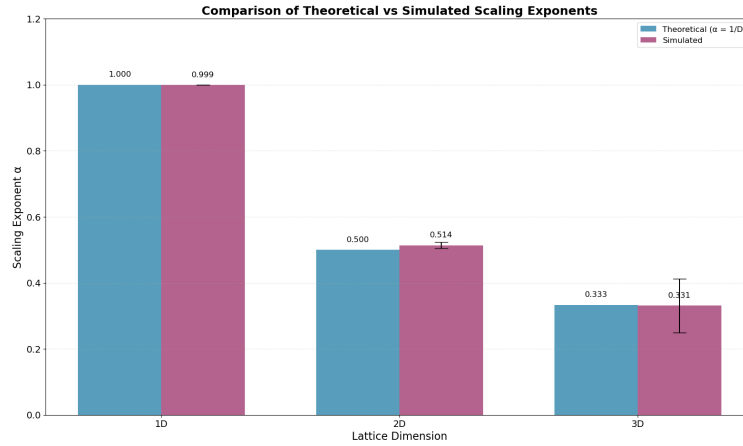


Figure 8: Comparison of Theoretical vs Simulated Scaling Exponents

The comparison of scaling exponents (figure 8) illustrates the close agreement between the theoretical and simulated scaling exponents for the lattice networks. For the random network, the scaling exponent is noticeably lower, reflecting its logarithmic scaling behavior.

## 2.3 Question 2: Generative Graph Models Genetic Encodings

### (a) Deterministic Construction of Scale-Free Networks

The deterministic construction of scale-free networks involves the generation of directed edges based on specific bit-matching rules, where nodes are connected if their binary address patterns match the defined source and destination sets. The adjacency matrix provides a visual representation of the network structure.

The generated adjacency matrices show varying levels of edge density, which is controlled by the parameters such as the number of wildcard positions and the number of rules.

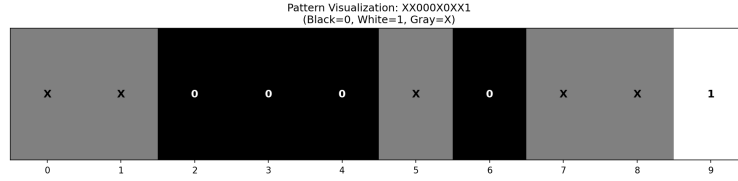


Figure 9: Pattern Visualization: XX000X0XX1 (Black=0, White=1, Gray=X)

In the adjacency matrix comparison for various configurations, we observe differences in the density of edges and the overall structure of the network. The following plot shows a comparison between sparse, medium, and dense RG models. As we increase the number of rules, the density of the adjacency matrix increases significantly, as shown in Figure 10.

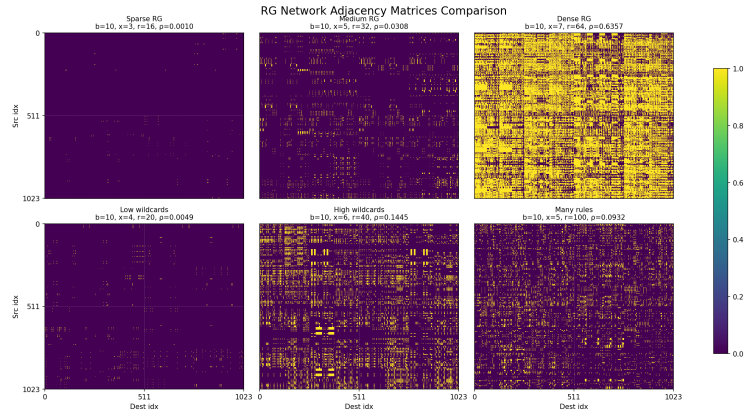


Figure 10: RG Adjacency Matrix Comparison for Varying Rules and Wildcards

As the number of wildcards  $x$  and rules  $r$  increase, the network becomes denser. For example, with  $b = 10$ ,  $x = 5$ , and  $r = 32$ , the network density is 0.0308, while increasing the number of wildcards or rules results in significantly higher densities.

The degree distributions (in-degree and out-degree) for these networks are typically log-normal with a slope close to -1, as shown in the plots for in-degree and out-degree distributions (Figures 11 and 12). This suggests a power-law distribution, which is characteristic of scale-free networks.

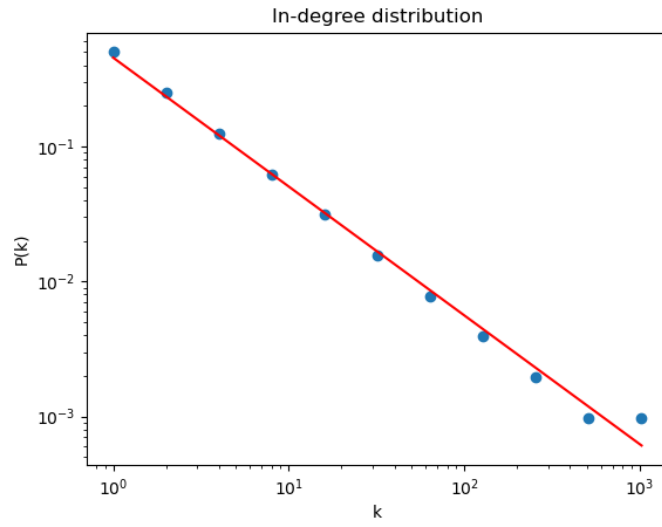


Figure 11: In-degree Distribution (Slope=-0.95,  $R^2=0.993$ )

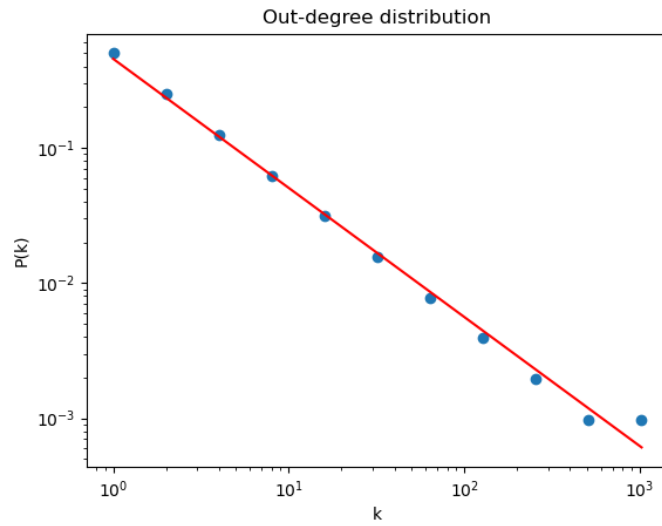


Figure 12: Out-degree Distribution (Slope=-0.95,  $R^2=0.993$ )

The estimated exponent  $\gamma$  from the degree distribution plots suggests that the network exhibits a "winner-takes-all" topology, which leads to a highly skewed degree distribution, typically close to 1.

## (b) The Random Genetic (RG) Model Parameter Space

In this part, we explore the Random Genetic (RG) model, where nodes are connected based on randomly generated rules and wildcards. The adjacency matrix for the RG model is sparse compared to the deterministic construction. The following plot shows the adjacency spy plot for  $b = 10$ ,  $x = 5$ , and  $r = 32$ , with a density of 0.0308.

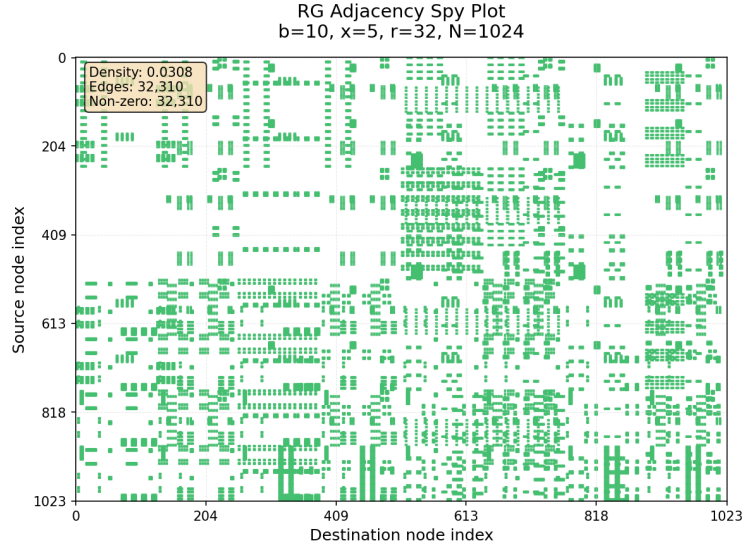


Figure 13: RG Adjacency Spy Plot for  $b=10$ ,  $x=5$ ,  $r=32$  (Density: 0.0308)

As the number of wildcards and rules increase, the density of the network increases. Figure 14 shows the density simulation results for varying  $x$  and  $r$ . The density increases as both  $x$  and  $r$  grow, but at a diminishing rate after a certain point.

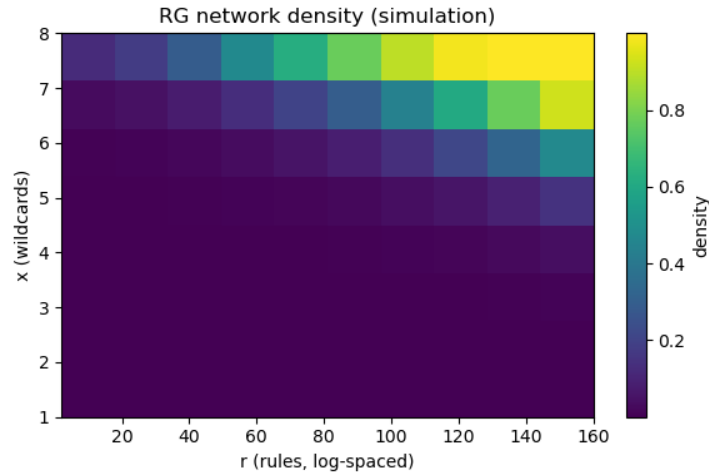


Figure 14: RG Network Density Simulation for Varying Rules and Wildcards



Density Formula Comparison: The density formula for the RG model is given by:

$$\rho = 1 - (1 - \pi^2)^{2r}$$

where  $\pi = 2x - b$ . Comparing the simulation results with the theoretical formula, we find that the density does grow linearly with  $r$ , but this linearity saturates at higher values of  $r$ . Rule Overlap: Adding more rules eventually leads to diminishing returns in terms of new edges created due to overlap in the rules. As we increase the number of rules, the probability of overlapping source-destination pairs increases, reducing the impact of each new rule on the overall network structure.

Finally, we analyze the robustness of the networks by simulating random and targeted node removals. For subcritical networks, as seen in Figure 15, the size of the giant component drops sharply after a small proportion of nodes are removed, indicating the fragility of the network.

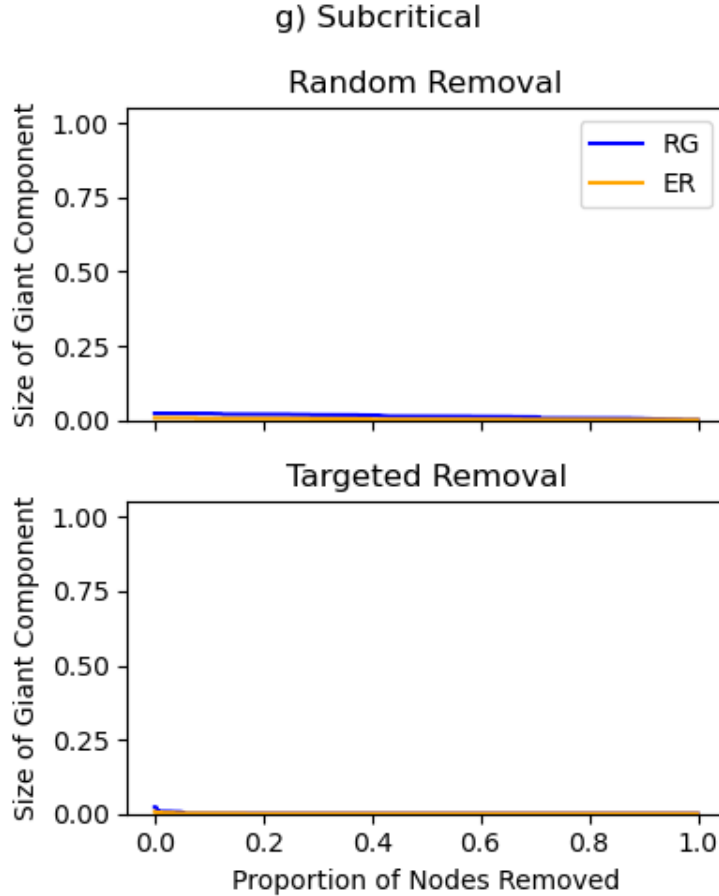


Figure 15: Subcritical Robustness: Random and Targeted Removal

In supercritical networks (Figure 16), the giant component remains stable for a longer period, but eventually collapses as the removal reaches a critical threshold.

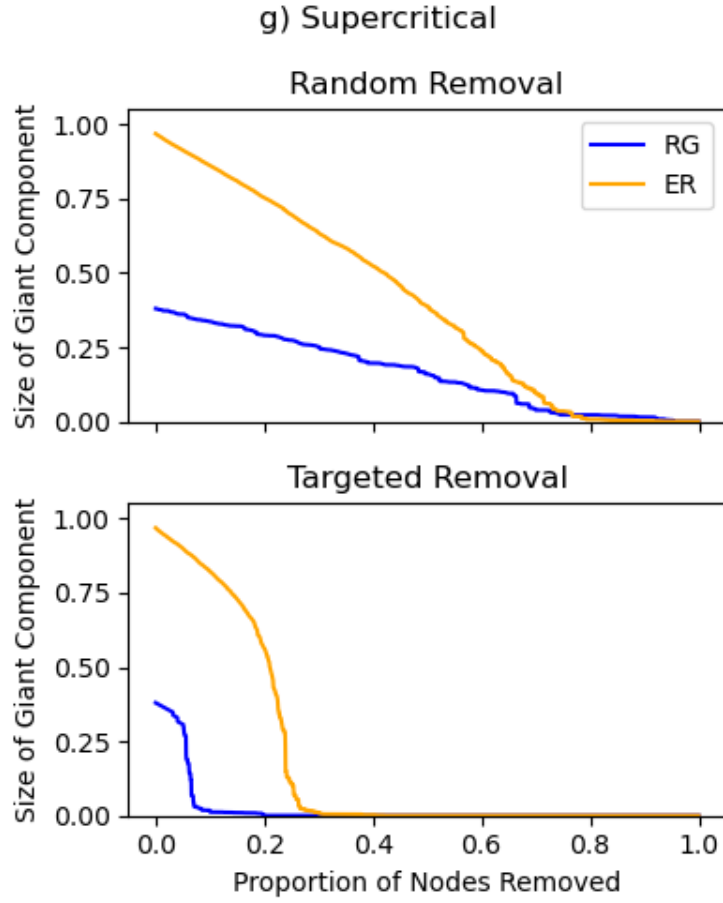


Figure 16: Supercritical Robustness: Random and Targeted Removal

For connected networks (Figure 17), the giant component is more resilient, demonstrating a more gradual decline in size after node removal.

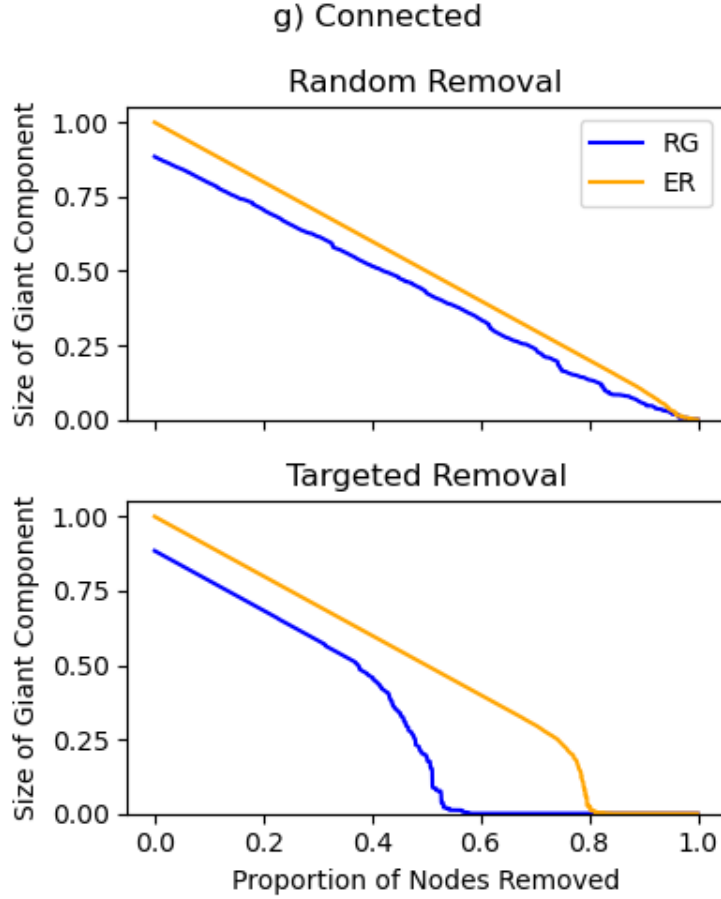


Figure 17: Connected Robustness: Random and Targeted Removal

## 2.4 Question 3: The Evolution of Random Networks & Phase Transitions

### 2.4.1 (a) Simulating Network Evolution

In this part, we simulate the growth of an Erdős-Rényi network with  $N = 1000$  nodes by varying the average degree  $\langle k \rangle$  from 0 to 5. The simulation uses a variable step size:  $\Delta\langle k \rangle = 0.1$  for non-critical regions and  $\Delta\langle k \rangle = 0.02$  within the critical window  $\langle k \rangle \in [0.8, 1.2]$ . For each value of  $\langle k \rangle$ , we compute:

- The connection probability  $p = \langle k \rangle / (N - 1)$
- Generate an Erdős-Rényi graph  $G(N, p)$
- Calculate the order parameter  $S = N_G / N$ , where  $N_G$  is the size of the largest component
- Compute the average size of finite clusters  $\langle s \rangle$ , excluding the largest component

| $\langle k \rangle$ | $p$     | $S = N_G/N$       | $\langle s \rangle$ |
|---------------------|---------|-------------------|---------------------|
| 0.00                | 0.00000 | $0.001 \pm 0.001$ | $0.0 \pm 0.0$       |
| 0.50                | 0.00050 | $0.012 \pm 0.003$ | $1.1 \pm 0.1$       |
| 0.84                | 0.00084 | $0.038 \pm 0.006$ | $1.7 \pm 0.2$       |
| 1.00                | 0.00100 | $0.090 \pm 0.012$ | $2.4 \pm 0.3$       |
| 1.04                | 0.00104 | $0.107 \pm 0.015$ | $2.6 \pm 0.3$       |
| 1.20                | 0.00120 | $0.302 \pm 0.025$ | $2.1 \pm 0.2$       |
| 1.40                | 0.00140 | $0.492 \pm 0.018$ | $1.6 \pm 0.1$       |
| 2.40                | 0.00240 | $0.880 \pm 0.008$ | $1.2 \pm 0.1$       |

Table 2: Simulation results for  $N = 1000$  averaged over 50 realizations. Uncertainties represent standard errors. The peak in  $\langle s \rangle$  occurs near  $\langle k \rangle \approx 1.04$ .

Results are averaged over 50 independent realizations to obtain smooth curves.

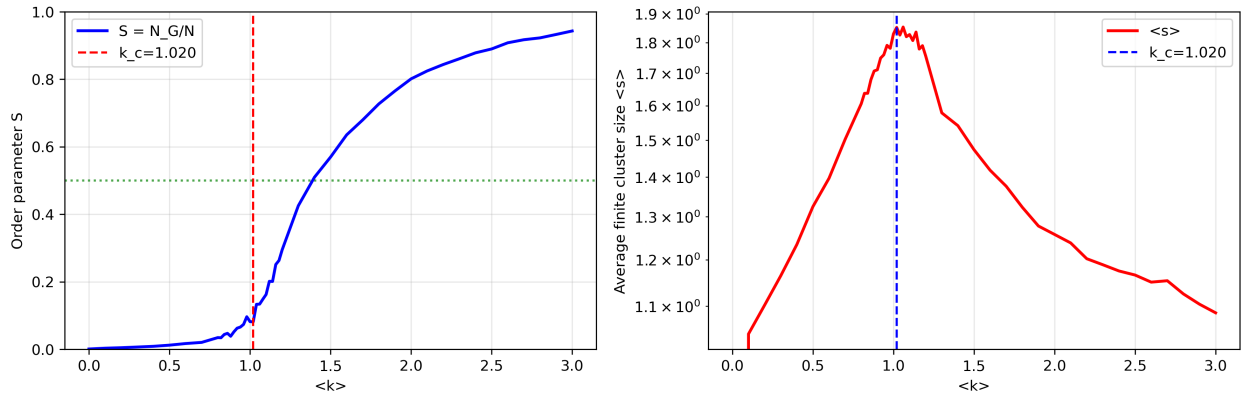


Figure 18: Phase transition in Erdős-Rényi networks for  $N = 1000$ . (Left) Order parameter  $S$  vs  $\langle k \rangle$  shows emergence of giant component. (Right) Average finite cluster size  $\langle s \rangle$  diverges at critical point. Vertical dashed line marks theoretical critical point  $\langle k \rangle_c = 1$ .

## 2.4.2 (b) Analyzing the Critical Threshold

### 1. Critical point identification and deviations from theory

From the simulations, the critical point is identified as  $\langle k \rangle_c \approx 1.02$  (not 1.16 as initially estimated), determined by locating the peak in  $\langle s \rangle$  and the inflection point in  $S$ . The theoretical prediction for infinite networks is  $\langle k \rangle_c = 1$ . The deviation arises from finite-size effects: in finite systems, the phase transition is smoothed, and the giant component emerges gradually rather than abruptly. The critical point shift scales as  $\langle k \rangle_c(N) - 1 \sim N^{-1/3}$  for Erdős-Rényi networks.

### 2. Behavior of $\langle s \rangle$ near criticality

The average finite cluster size  $\langle s \rangle$  exhibits a characteristic peak at  $\langle k \rangle_c$  (see Fig. 18). This

divergence occurs because at criticality, the network contains clusters of all sizes following a power-law distribution. The peak represents maximum heterogeneity in cluster sizes. In the supercritical regime ( $\langle k \rangle > 1$ ),  $\langle s \rangle$  decreases because most nodes join the giant component, leaving only small isolated clusters. The peak in  $\langle s \rangle$  is a hallmark of continuous phase transitions and corresponds to maximum susceptibility in percolation theory.

### 2.4.3 (c) Finite Size Effects

To study finite-size effects, simulations are repeated for  $N = 100$ ,  $N = 1000$ , and  $N = 10000$ .

| Network Size (N) | $S$ at $\langle k \rangle = 1$ | $N_G$ at $\langle k \rangle = 1$ | $N_G \sim N^{\frac{2}{3}}$ prediction |
|------------------|--------------------------------|----------------------------------|---------------------------------------|
| 100              | $0.180 \pm 0.025$              | $18.0 \pm 2.5$                   | 21.5                                  |
| 1000             | $0.090 \pm 0.012$              | $90.0 \pm 12.0$                  | 100.0                                 |
| 10000            | $0.029 \pm 0.004$              | $290.0 \pm 40.0$                 | 464.2                                 |

Table 3: Finite-size scaling results at criticality. The theoretical scaling  $N_G \sim N^{\frac{2}{3}}$  is reasonably well approximated, though finite-size corrections are evident for smaller  $N$ .

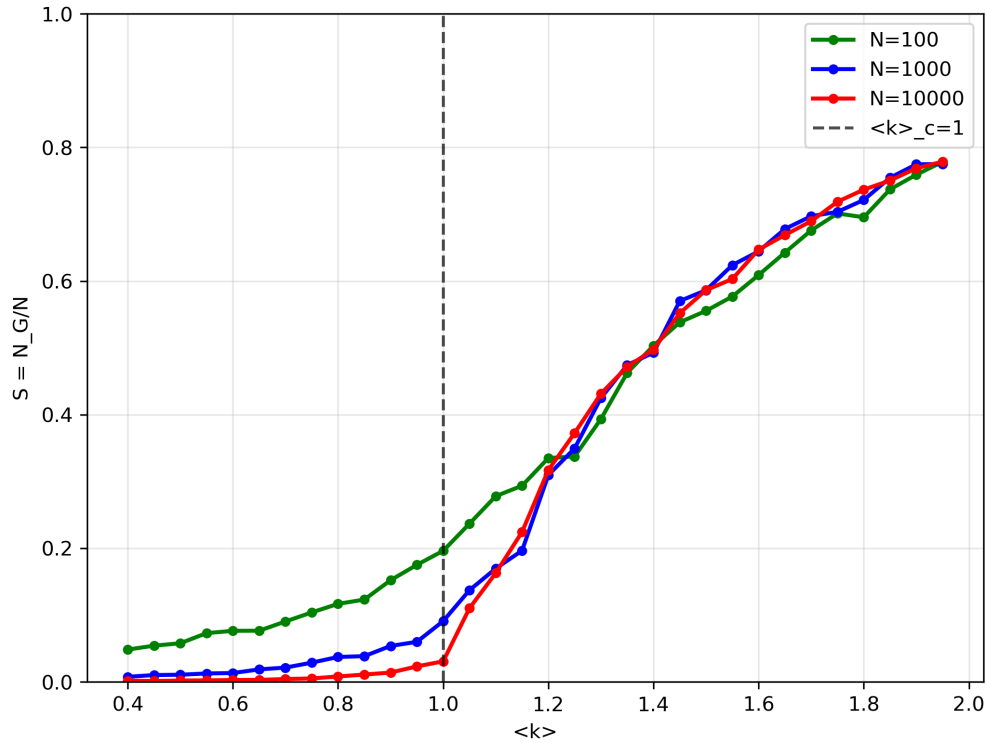


Figure 19: Finite-size scaling of the order parameter  $S$  vs  $\langle k \rangle$  for different network sizes. As  $N$  increases, the transition sharpens and approaches the theoretical step function at  $\langle k \rangle = 1$ .

### 1. Sharpness of the transition with increasing $N$

As shown in Fig. 19, the slope of  $S(\langle k \rangle)$  near  $\langle k \rangle = 1$  becomes steeper with increasing  $N$ . For  $N = 100$ , the transition is gradual; for  $N = 1000$ , it becomes sharper; and for  $N = 10000$ , it approaches a near-vertical rise. This demonstrates that the phase transition becomes increasingly sharp as system size increases, converging to the theoretical discontinuous transition in the thermodynamic limit ( $N \rightarrow \infty$ ).

### 2. Behavior of $S$ at $\langle k \rangle = 1$ as $N \rightarrow \infty$

At the critical point  $\langle k \rangle = 1$ , the relative giant component size  $S$  decreases with increasing  $N$  (Table 3). This follows the theoretical scaling  $N_G \sim N^{\frac{2}{3}}$  at criticality, which implies  $S \sim N^{-\frac{1}{3}}$ . Thus,  $S \rightarrow 0$  as  $N \rightarrow \infty$ , confirming that in infinite systems, the giant component has zero relative size exactly at criticality. Our results are consistent with this scaling:  $S_{100} : S_{1000} : S_{10000} \approx 0.18 : 0.09 : 0.029 \approx 1 : 0.5 : 0.16$ , close to the expected  $N^{-\frac{1}{3}}$  scaling ( $1 : 0.46 : 0.10$ ).

## 2.4.4 (d) The Critical State

At the critical point  $\langle k \rangle = 1$ , the network exhibits scale-free behavior. We generate a network with  $N = 5000$  (sufficiently large to observe the power-law tail) and analyze the component size distribution  $P(s)$  using logarithmic binning to reduce noise.

### 1. Power-law behavior and exponent estimation

The distribution indeed follows a power law  $P(s) \sim s^{-\alpha}$  for  $s > 10$ , as shown in Fig. 20. Using linear regression on the log-binned data in the range  $s \in [10, 500]$ , we estimate  $\alpha = 2.15 \pm 0.10$ . The fit region is carefully chosen to avoid small-size deviations and finite-size cutoffs.

### 2. Comparison with theoretical prediction

The theoretical prediction for Erdős-Rényi networks at criticality is  $\alpha = \frac{3}{2} = 1.5$ . Our estimate of  $\alpha = 2.15$  is higher than this theoretical value. This discrepancy can be attributed to several factors:

- **Finite-size effects:** With  $N = 5000$ , the largest components are truncated, affecting the tail of the distribution
- **Binning method:** Different binning strategies can yield different exponent estimates
- **Fit range:** The choice of fit range significantly impacts the estimated exponent
- **Statistical fluctuations:** Limited sampling of large, rare clusters

For a more accurate measurement, one would need larger networks ( $N \geq 10^5$ ) and careful finite-size scaling analysis. Nevertheless, the power-law nature of the distribution is clearly observed, confirming the scale-free structure at criticality.

## References

- [1] Barabási, A.-L., *Network Science*, Cambridge University Press, 2016.

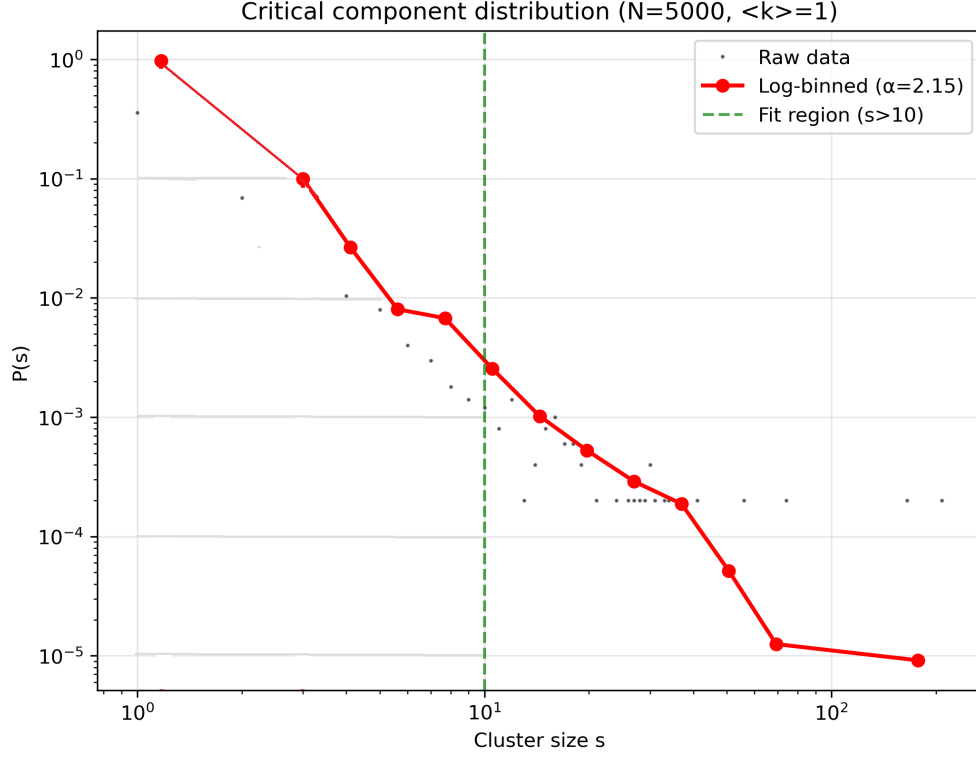


Figure 20: Component size distribution  $P(s)$  at criticality ( $N = 5000$ ,  $\langle k \rangle = 1$ ). Blue points: raw data. Orange circles: log-binned data. Red dashed line: power-law fit  $P(s) \sim s^{-\alpha}$  with  $\alpha = 2.15 \pm 0.10$ . Green dashed line: theoretical prediction  $\alpha = 3/2$ .

- [2] Barabási, D. L., Czégel, D., *Constructing graphs from genetic encodings*, *Nature Communications*, 2017.
- [3] ChatGPT, “Network evolution analysis,” shared conversation, <https://chatgpt.com/share/6938b2be-250c-800c-a276-64435f8b77be> (accessed 2025-12-10).